The whatsnote Class*

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Released 2025-11-12 v5.0B

1 Introduction

The whatsnote LaTeX class provides an elegant layout and powerful tools for taking notes in science, engineering, etc. It supports various compilation methods, such as pdfLaTeX, XeLaTeX, ApLaTeX, LuaLaTeX, etc.

2 Usage

To load this class, write the line

```
\documentclass{whatsnote}
```

2.1 Cover configuration

`\coverset` `\coverset {⟨keys⟩}`

The `\coverset` macro can config the cover, the mandatory argument accepts the following keys

title = ⟨*string*⟩ can set the title on the cover page.

author, **afill** = ⟨*string*⟩ can set the author and the affliction respectively on the cover page, and they will be combined together with a vertical bar separated.

date = ⟨*string*⟩ can set the date on the cover page.

extinfo = ⟨*string*⟩ can add contents at the south east corner of the cover page (e.g., add your personal seal ("Yìn Zhāng" in Chinese)).

head, **clogo** = ⟨*file*⟩ can add pictures at the top / the center white area of the cover page accordingly.

llogos = ⟨*files' comma list*⟩ can add a series of pictures at the "left bar" of the title page.

`\maketitle` `\maketitle [⟨color⟩]`

The `\maketitle` macro can generate the cover and config the theme color in its optional argument.

*https://github.com/myhsia/whatsnote, https://ctan.org/pkg/whatsnote

\sidehead \sidehead {⟨*sidebar header*⟩} \sidefoot {⟨*sidebar footer*⟩}
\sidefoot
The \sidehead, sidefoot macros can config the header and footer in the sidebar.

\mathemph \mathemph {⟨*math equation*⟩}

The \mathemph macro can color-emphing math formula.

3 Working Example

```
\documentclass[11pt]{whatsnote}

\usepackage[versionfa = 7, hook]{fontawesomescaled}
\usepackage{duckuments}
\coverset{
  title   = Demo for the \textsc{WhatsNote} Package,
  author  = \ttfamily \textsc{Mingyu Xia},
  afill   = <\href{mailto:xiamingyu@westlake.edu.cn}
              {xiamingyu@westlake.edu.cn}>,
  date    = November 2025,
  extinfo = {\href{https://github.com/myhsia/whatsnote}
              {\faIconScaled[scale = 4]{github}}},
  head    = example-image-golden.pdf,
  clogo   = example-grid-100x100pt.pdf,
  llogos  = {example-image-duck, example-image-duck,
             example-image-duck, example-image-duck}
}

\begin{document}

\maketitle[teal]
\frontmatter \tableofcontents \mainmatter

\duckument[full]

\end{document}
```

Golden ratio

(Original size: 32.361×200 bp)

Demo for the WhatsNote Package

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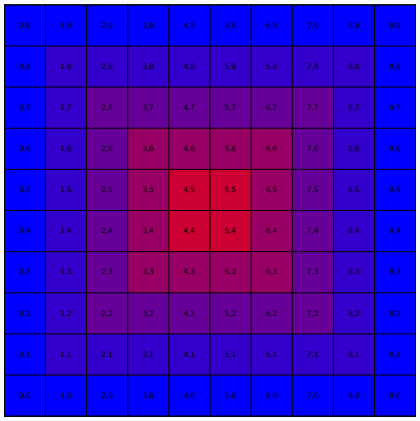

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Chapter 1 A friendly duck at level 0 (chapter)

There once was a very smart but sadly blind duck. When it was still a small duckling it was renowned for its good vision. But sadly as the duck grew older it caught a sickness which caused its eyesight to worsen. It became so bad, that the duck couldn't read the notes it once took containing much of inline math. Only displayed equations remained legible. That annoyed the smart duck, as it wasn't able to do its research any longer. It called for its underduckling and said: "Go, find me the best eye ducktor there is. He shall heal me from my disease!"

"But my duck, how are you supposed to manage your daily routines without my visual guidance", replied the underduckling. The smart duck's face turned grim in anger. "You dare to talk back?" The underduckling blushed ashamed. How could he have objections after his duck gave strict orders? The underduckling was so embarrassed about his own behaviour he had to solve an equation.

After the equation was solved and the underduckling prepared his leave for the next day it fell asleep in a shaky mood. It did not know what the journey had prepared for him and if he was prepared enough for it. His sleep was restless. The dreams he had that night were not calm and bright as they used to be for an innocent underduckling.

Before dawn the underduckling woke. He didn't have the feeling of being well rested. But nonetheless he knew that this was the day he should leave. Except saying goodbye to his beloved ones there was nothing holding him back. His duck had sent him on the most important mission a five weeks old inexperienced underduckling was ever sent on. He bid farewell to his mother, all his brothers and sisters, and finally from his duck. The bag was shouldered, the boots were tied, the underduckling left.

As the underduckling went along he saw a tree. It was so tall he couldn't even approximate its height. Oh, how small he felt. What should he, a small duckling whose plumage was still fluffy, achieve in this vast, gigantic world? He stuck his trembling wings under the straps of his backpack to calm his shaking limbs. Where should he start his search? How should he survive in the hostile nature surrounding the sheltered pond of his community?

1.1 A friendly duck at level 1 (section)

There once was a very smart but sadly blind duck. When it was still a small duckling it was renowned for its good vision. But sadly as the duck grew older it caught a sickness which caused its eyesight to worsen. It became so bad, that the duck couldn't read the notes it once took containing much of inline math. Only displayed equations remained legible. That annoyed the smart duck, as it wasn't able to do its research any longer. It called for its underduckling and said: "Go, find me the best eye ducktor there is. He shall heal me from my disease!"

"But my duck, how are you supposed to manage your daily routines without my visual guidance", replied the underduckling. The smart duck's face turned grim in anger. "You dare to talk back?" The underduckling

blushed ashamed. How could he have objections after his duck gave strict orders? The underduckling was so embarrassed about his own behaviour he had to solve an equation.
After the equation was solved and the underduckling prepared his leave for the next day it fell asleep in a shaky mood. It did not know what the journey had prepared for him and if he was prepared enough for it. His sleep was restless. The dreams he had that night were not calm and bright as they used to be for an innocent underduckling.
Before dawn the underduckling woke. He didn't have the feeling of being well rested. But nonetheless he knew that this was the day he should leave. Except saying goodbye to his beloved ones there was nothing holding him back. His duck had sent him on the most important mission a five weeks old inexperienced underduckling was ever sent on. He bid farewell to his mother, all his brothers and sisters, and finally from his duck. The bag was shouldered, the boots were tied, the underduckling left.
As the underduckling went along he saw a tree. It was so tall he couldn't even approximate its height. Oh, how small he felt. What should he, a small duckling whose plumage was still fluffy, achieve in this vast, gigantic world? He stuck his trembling wings under the straps of his backpack to calm his shaking limbs. Where should he start his search? How should he survive in the hostile nature surrounding the sheltered pond of his community?

1.1.1 A friendly duck at level 2 (subsection)

There once was a very smart but sadly blind duck. When it was still a small duckling it was renowned for its good vision. But sadly as the duck grew older it caught a sickness which caused its eyesight to worsen. It became so bad, that the duck couldn't read the notes it once took containing much of inline math. Only displayed equations remained legible. That annoyed the smart duck, as it wasn't able to do its research any longer. It called for its underduckling and said: "Go, find me the best eye ducktor there is. He shall heal me from my disease!"
"But my duck, how are you supposed to manage your daily routines without my visual guidance", replied the underduckling. The smart duck's face turned grim in anger. "You dare to talk back?" The underduckling blushed ashamed. How could he have objections after his duck gave strict orders? The underduckling was so embarrassed about his own behaviour he had to solve an equation.
After the equation was solved and the underduckling prepared his leave for the next day it fell asleep in a shaky mood. It did not know what the journey had prepared for him and if he was prepared enough for it. His sleep was restless. The dreams he had that night were not calm and bright as they used to be for an innocent underduckling.
Before dawn the underduckling woke. He didn't have the feeling of being well rested. But nonetheless he knew that this was the day he should leave. Except saying goodbye to his beloved ones there was nothing holding him back. His duck had sent him on the most important mission a five weeks old inexperienced underduckling was ever sent on. He bid farewell to his mother, all his brothers and sisters, and finally from his duck. The bag was shouldered, the boots were tied, the underduckling left.
As the underduckling went along he saw a tree. It was so tall he couldn't even approximate its height. Oh, how small he felt. What should he, a small duckling whose plumage was still fluffy, achieve in this vast,

Demo for the WhatsNote Package

gigantic world? He stuck his trembling wings under the straps of his backpack to calm his shaking limbs. Where should he start his search? How should he survive in the hostile nature surrounding the sheltered pond of his community?

A friendly duck at level 3 (subsubsection)

There once was a very smart but sadly blind duck. When it was still a small duckling it was renowned for its good vision. But sadly as the duck grew older it caught a sickness which caused its eyesight to worsen. It became so bad, that the duck couldn't read the notes it once took containing much of inline math. Only displayed equations remained legible. That annoyed the smart duck, as it wasn't able to do its research any longer. It called for its underduckling and said: "Go, find me the best eye ducktor there is. He shall heal me from my disease!"
"But my duck, how are you supposed to manage your daily routines without my visual guidance", replied the underduckling. The smart duck's face turned grim in anger. "You dare to talk back?" The underduckling blushed ashamed. How could he have objections after his duck gave strict orders? The underduckling was so embarrassed about his own behaviour he had to solve an equation.
After the equation was solved and the underduckling prepared his leave for the next day it fell asleep in a shaky mood. It did not know what the journey had prepared for him and if he was prepared enough for it. His sleep was restless. The dreams he had that night were not calm and bright as they used to be for an innocent underduckling.
Before dawn the underduckling woke. He didn't have the feeling of being well rested. But nonetheless he knew that this was the day he should leave. Except saying goodbye to his beloved ones there was nothing holding him back. His duck had sent him on the most important mission a five weeks old inexperienced underduckling was ever sent on. He bid farewell to his mother, all his brothers and sisters, and finally from his duck. The bag was shouldered, the boots were tied, the underduckling left.
As the underduckling went along he saw a tree. It was so tall he couldn't even approximate its height. Oh, how small he felt. What should he, a small duckling whose plumage was still fluffy, achieve in this vast, gigantic world? He stuck his trembling wings under the straps of his backpack to calm his shaking limbs. Where should he start his search? How should he survive in the hostile nature surrounding the sheltered pond of his community?

A friendly duck at level 4 (paragraph) There once was a very smart but sadly blind duck. When it was still a small duckling it was renowned for its good vision. But sadly as the duck grew older it caught a sickness which caused its eyesight to worsen. It became so bad, that the duck couldn't read the notes it once took containing much of inline math. Only displayed equations remained legible. That annoyed the smart duck, as it wasn't able to do its research any longer. It called for its underduckling and said: "Go, find me the best eye ducktor there is. He shall heal me from my disease!"
"But my duck, how are you supposed to manage your daily routines without my visual guidance", replied the underduckling. The smart duck's face turned grim in anger. "You dare to talk back?" The underduckling blushed ashamed. How could he have objections after his duck gave strict orders? The underduckling was so embarrassed about his own behaviour he had to solve an equation.

Demo for the WhatsNote Package
Demo for the WhatsNote Package

After the equation was solved and the underduckling prepared his leave for the next day it fell asleep in a shaky mood. It did not know what the journey had prepared for him and if he was prepared enough for it. His sleep was restless. The dreams he had that night were not calm and bright as they used to be for an innocent underduckling.
Before dawn the underduckling woke. He didn't have the feeling of being well rested. But nonetheless he knew that this was the day he should leave. Except saying goodbye to his beloved ones there was nothing holding him back. His duck had sent him on the most important mission a five weeks old inexperienced underduckling was ever sent on. He bid farewell to his mother, all his brothers and sisters, and finally from his duck. The bag was shouldered, the boots were tied, the underduckling left.
As the underduckling went along he saw a tree. It was so tall he couldn't even approximate its height. Oh, how small he felt. What should he, a small duckling whose plumage was still fluffy, achieve in this vast, gigantic world? He stuck his trembling wings under the straps of his backpack to calm his shaking limbs. Where should he start his search? How should he survive in the hostile nature surrounding the sheltered pond of his community?

1.2 Lists

1.2.1 Example for ducks (itemize)

- First swims father drake
- Then floats mother duck
- After her paddles baby duckling
- And over there bathes uncle canard

Nested ducks

- swimming father drake
  - swimming father drake
    - swimming father drake
      - swimming father drake
      - floating mother duck
    - floating mother duck
  - floating mother duck
- floating mother duck

1.2.2 Example for ducks (enumerate)

1. First swims father drake
2. Then floats mother duck
3. After her paddles baby duckling
4. And over there bathes uncle canard

Nested ducks

1. swimming father drake
   (a) swimming father drake
      i. swimming father drake
         A. swimming father drake
         B. floating mother duck
      ii. floating mother duck
   (b) floating mother duck
2. floating mother duck

1.2.3 Example for ducks (description)

drake is the swimming father

duck is the floating mother

duckling is the paddling baby

canard is the bathing uncle

Nested ducks

drake is the swimming father

drake is the swimming father
drake is the swimming father
drake is the swimming father
drake is the swimming father
duck is the floating mother
duck is the floating mother
duck is the floating mother

duck is the floating mother

A The Source Code

Assign the module name of the variables and control sequences, which will be automatically replaced by l3docstrip.

```
⟨@@=whats⟩
```

A.1 The `whatsnote.cls` file

Start the optionlist `class` for l3docstrip.

```
⟨*class⟩
```

Define the date, version, and summary variables, and declare the class.

```
\def \whatsnote@date    {2025-11-12}
\def \whatsnote@version {v5.0B}
\def \whatsnote@tldr
  {LaTeX class for taking notes in science, engineering, etc.}
\ProvidesExplClass {whatsnote} {\whatsnote@date}
  {\whatsnote@version} {\whatsnote@tldr}
```

`\__whats_msg_new:nn` `\__whats_msg_error:nn` `\__whats_msg_warning:nn`

Functions for broadcast errors and warnings.

```
\cs_new_protected:Npn \__whats_msg_new:nn #1#2
  { \msg_new:nnn { whatsnote } {#1} {#2} }
\cs_new_protected:Npn \__whats_msg_error:nn #1#2
  { \msg_error:nnn { whatsnote } {#1} {#2} }
\cs_new_protected:Npn \__whats_msg_warning:nn #1#2
  { \msg_error:nnn { whatsnote } {#1} {#2} }
```

(*End of definition for* `\__whats_msg_new:nn`, `\__whats_msg_error:nn`, *and* `\__whats_msg_warning:nn`.)

`\__whats_provide_module:n`

Function for providing modules with the predefined date and version variables.

```
\cs_new_protected_nopar:Npn \__whats_provide_module:n #1
  {
    \ProvidesExplFile {whats-#1-module.code.tex}
      {\whatsnote@date} {\whatsnote@version}
      { whatsNote ~ \text_titlecase:n {#1} ~ Module }
  }
```

Function for loading modules.

`\__whats_load_module:n`

```
\cs_new_protected_nopar:Npn \__whats_load_module:n #1
  {
    \clist_map_inline:nn {#1}
      {
        \file_if_exist_input:nF { whats-##1-module.code.tex }
          { \__whats_msg_error:nn { not found module } {##1} }
      }
  }
\__whats_msg_new:nn { not found module }
  { The ~ whats ~ module ~ ‘#1’ ~ not ~ found. }
```

(*End of definition for* `\__whats_provide_module:n` *and* `\__whats_load_module:n`.)

\g__whats_base_class_options_clist Variable for storing global option for basic class.

```
\clist_new:N \g__whats_base_class_options_clist
\keys_define:nn { whats / classoption }
  {
    unknown .code:n = \__whats_unknown_option:n { #1 },
  }
```

(*End of definition for* `\g__whats_base_class_options_clist`.)

__whats_unknown_option:n Parsing unknown options: pass them to the basic class.

```
\cs_new_protected_nopar:Npn \__whats_unknown_option:n #1
  {
    \tl_if_empty:nTF { #1 }
      { \clist_gput_right:NV \g__whats_base_class_options_clist \l_keys_key_str }
      {
        \exp_args:NNx \clist_gput_right:Nn \g__whats_base_class_options_clist
          { \l_keys_key_str = \exp_not:n {#1} }
      }
  }
\ProcessKeyOptions [ whats / classoption ]
```

(*End of definition for* `\__whats_unknown_option:n`.)

Load basic class with some preloaded options.

```
\PassOptionsToClass { book } { 11pt }
\PassOptionsToPackage { svgnames } { xcolor }
\exp_args:NNV \LoadClass [ \g__whats_base_class_options_clist ] { book }
```

Load modules for whatsnote.

```
\__whats_load_module:n { typeset, layout, theorem, cover }
```

End the input stream of `whatsnote.cls`.

```
\file_input_stop:
```

End the optionlist `class` for l3docstrip.

```
⟨/class⟩
```

A.2 The `whats-typeset-module.code.tex` file

Start the optionlist `typeset` for l3docstrip.

```
⟨*typeset⟩
```

Declare the module file.

```
\__whats_provide_module:n { typeset }
```

Load the amssymb, bbm, bm, cancel, extarrows, mathtools, nicematrix, xfrac packages.

```
\RequirePackage{amssymb, bbm, bm, cancel, extarrows,
                mathtools, nicematrix, xfrac}
```

Load the `e` style esvect package for `\vec` arrow, then define the anti-vector arrow.

```
\RequirePackage[e]{esvect}
\let \vec \vv
\DeclareRobustCommand \cev [1]{{\mathpalette\do@cev{#1}}}
\newcommand \do@cev [2]{
  \vbox{\offinterlineskip
    \ialign{##\cr
      \hidewidth\reflectbox{$\m@th#1\vec{}\mkern4mu$}\hidewidth\cr
```

```
      $\m@th#1#2$\cr
    }
  }
}
```

Load the fixdif and derivative packages for inputting math derivatives, suppress warnings when using `xetex`.

```
\sys_if_engine_xetex:T { \PassOptionsToPackage { no-math } { fontspec } }
\RequirePackage{fixdif, derivative}
```

Define some mathematic operators.

```
\newdif \D {\mathsf D}
\newdif \upe {\mathrm e}
\newdif \iu {\mathrm i \mkern1mu}
\let \sinh \relax \let \cosh \relax \let \tanh \relax \let \coth \relax
\DeclareMathOperator \sinh  {sh}
\DeclareMathOperator \cosh  {ch}
\DeclareMathOperator \tanh  {th}
\DeclareMathOperator \coth  {cth}
\DeclareMathOperator \sgn   {sgn}
\DeclareMathOperator \cotan {\qopname\relax o{ctg}}
\DeclareMathOperator \im    {im}
\DeclareMathOperator \adj   {adj}
\DeclareRobustCommand \tran{^{\mkern-1.5mu\mathsf T}}
\DeclareRobustCommand \identity {\ensuremath{\mathbbm 1}}
```

Load the siunitx package for inputting physics units, then declare the unit “angstrom”.

```
\RequirePackage{siunitx}
\DeclareSIUnit \angstrom {\text \AA}
```

Load the mhchem package for inputting chemistry formula.

```
\RequirePackage[version = 4]{mhchem}
```

Load the physics2 package's some modules for quick math and physics input. Temporarily change the module name for hacking the package[1].

```
⟨@@=phy⟩
\RequirePackage{physics2}
\usephysicsmodule{ab, ab.braket, diagmat, xmat, qtext.legacy, op.legacy}
```

Switch the star and non-star branches for the Dirac brackets inputting commands to make the non-equal height brackets the default.

```
\DeclareRobustCommand \bra
  {\@ifstar{\phy@d@lx{br.m}{br.a}}{\phy@d@lx{br.m}{br.a}*}}
\DeclareRobustCommand \ket
  {\@ifstar{\phy@d@lx{kt.m}{kt.a}}{\phy@d@lx{kt.m}{kt.a}*}}
\DeclareRobustCommand \braket
  {\@ifstar{\phy@d@lx{bk.m}{bk.a}}{\phy@d@lx{bk.m}{bk.a}*}}
\DeclareRobustCommand \ketbra
  {\@ifstar{\phy@d@lx{kb.m}{kb.a}}{\phy@d@lx{kb.m}{kb.a}*}}
```

Extend the `diagmat` module for the physics2 package: support the small matrices, and enable users to modify the alignments of the matrices' elements.

```
\DeclareDocumentCommand \diagmat { s O{} m }
  {
    \IfBooleanTF {#1}
```

[1] `@AlphaZTX` has not merged my suggestion for the physics2 package. Blame him!

```
      { \__phy_diagmat_type:nnn {  small } {#2} {#3} }
      { \__phy_diagmat_type:nnn {        } {#2} {#3} }
  }
\DeclareDocumentCommand \pdiagmat { s O{} m }
  {
    \IfBooleanTF {#1}
      { \__phy_diagmat_type:nnn { psmall } {#2} {#3} }
      { \__phy_diagmat_type:nnn { p      } {#2} {#3} }
  }
\DeclareDocumentCommand \bdiagmat { s O{} m }
  {
    \IfBooleanTF {#1}
      { \__phy_diagmat_type:nnn { bsmall } {#2} {#3} }
      { \__phy_diagmat_type:nnn { b      } {#2} {#3} }
  }
\DeclareDocumentCommand \Bdiagmat { s O{} m }
  {
    \IfBooleanTF {#1}
      { \__phy_diagmat_type:nnn { Bsmall } {#2} {#3} }
      { \__phy_diagmat_type:nnn { B      } {#2} {#3} }
  }
\DeclareDocumentCommand \vdiagmat { s O{} m }
  {
    \IfBooleanTF {#1}
      { \__phy_diagmat_type:nnn { vsmall } {#2} {#3} }
      { \__phy_diagmat_type:nnn { v      } {#2} {#3} }
  }
\DeclareDocumentCommand \Vdiagmat { s O{} m }
  {
    \IfBooleanTF {#1}
      { \__phy_diagmat_type:nnn { Vsmall } {#2} {#3} }
      { \__phy_diagmat_type:nnn { V      } {#2} {#3} }
  }
```

Restore the module name.

```
⟨@@=whats⟩
```

Load and config the hyperref package, suppress the bookmark warnings.

```
\RequirePackage{hyperref}
\hypersetup{colorlinks, citecolor = teal}
\let\oldHyPsd@CatcodeWarning\HyPsd@CatcodeWarning
\renewcommand \HyPsd@CatcodeWarning [1]
  {\ifnum\pdfstrcmp{#1}{math shift}=0 \else \oldHyPsd@CatcodeWarning{#1} \fi}
\pdfstringdefDisableCommands{
  \def\intop{int} \def\ilimits@{lim}
  \def\infty{infty} \let\HyPsd@CatcodeWarning\@gobble
}
```

\mathemph Macro for color-emphing math formula.

```
\NewDocumentCommand \mathemph { O{} m } {\ifmmode
  \colorbox{Crimson!15}{\vphantom{$\displaystyle #1$}$\displaystyle #2$}\else
  \colorbox{Crimson!15}{\vphantom{#1}#2}\fi}
```

(*End of definition for* \mathemph. *This function is documented on page* 2.)

```
\color_set:nnn { eqcolor } { HTML } { 2F4F4F }
```

Restore the definition for \[and \] changed by amsmath, then apply the equations' color (high-unstable, temporarily disabled).

```
% \def \dollardollar@begin { \def \eqcolor {\color{eqcolor}} \eqcolor $$ }
% \def \dollardollar@end   { $$ \color_select:n {black} }
\DeclareRobustCommand \[
  {
    \scan_stop:
    \if_mode_math:
      \@badmath
    \else:
      \if_mode_vertical:
        \nointerlineskip
        \makebox[.6\linewidth]{ }
      \fi:
    \dollardollar@begin % \color_select:n { eqcolor }
    \fi:
  }
\DeclareRobustCommand \]
  {
    \scan_stop:
    \if_mode_math:
      \if_mode_inner:
        \@badmath
      \else:
        \dollardollar@end
      \fi:
    \else:
      \@badmath
    \fi:
    \ignorespaces
  }
```

End the input stream of `whats-typeset-module.code.tex`.

```
\file_input_stop:
```

End the optionlist `typeset` for l3docstrip.

```
⟨/typeset⟩
```

A.3 The `whats-layout-module.code.tex` file

Start the optionlist `layout` for l3docstrip.

```
⟨*layout⟩
```

Declare the module file.

```
\__whats_provide_module:n { layout }
```

Load the anyfontsize, lmodern, microtype, manfnt, ragged2e, soul, libertine, ulem, xcolor packages.

```
\RequirePackage{anyfontsize, lmodern, microtype, manfnt, ragged2e, soul}
\RequirePackage[mono = false, osf]{libertine}
\RequirePackage[normalem]{ulem}
\RequirePackage[svgnames]{xcolor}
```

Config the geometry.

```
\RequirePackage[left  = 1.25in, vmargin = 1in, headheight = 15pt,
                right = .75in,  headsep = 15pt]{geometry}
```

```
\RequirePackage{graphicx, pgfplots}
\graphicspath{{./media/logos/} {./media/}}
\pgfplotsset{compat = newest}
% Table and list
\RequirePackage{booktabs, array, enumext}
\setenumext[enumext, 1]{label = (\alph*)}
\setenumext[enumext, 2]{label = \roman*.}
\RequirePackage[inline]{enumitem}
```

Config the header and footer.

```
\RequirePackage{fancyhdr}
\renewcommand* \chaptermark [1]
  {
    \markboth
      {
        \normalfont \biolinumLF \scshape
        \int_compare:nNnT { \c@secnumdepth } > { \m@ne }
          { \if@mainmatter \@chapapp \ \thechapter .~\ \fi } #1
      } {}
  }
\renewcommand* \sectionmark [1]
  {
    \markright
      {
        \normalfont \sffamily
        \int_compare:nNnT { \c@secnumdepth } > { \z@ } { \thesection . \ } #1
      }
  }
\fancyhead[OR]{\sffamily \rightmark}
\fancyhead[OL]{\sffamily \thepage}
\fancyhead[EL]{\sffamily \leftmark}
\fancyhead[ER]{\sffamily \thepage}
\fancyfoot[L,C,R]{}
\pagestyle{fancy}
```

Config the header rule and footer rule style.

```
\renewcommand* \headrulewidth {1pt}
\renewcommand* \footrulewidth {.5pt}
\hook_gput_code:nnn { cmd / headrule / before } {.}
  { \color_select:n { black!25 } }
\renewcommand \footrule
  {
    \if@fancyplain \let \footrulewidth \plainfootrulewidth \fi
    \color_select:n { black!25 }
    \hrule \@width \headwidth \@height \footrulewidth
  }
```

Paragraphs configurations.

```
\linespread{1.2}
\RequirePackage{titlesec, subcaption}
\titleformat \chapter [hang] {\huge \bfseries \sffamily}
  {\biolinumLF \scshape \@chapapp\ \thechapter} {1em} {}
\titlespacing* \chapter {\z@} {\z@} {30pt}
\titlespacing \paragraph {\z@} {1.5ex minus .1 ex} {1pc}
\RequirePackage{paracol}
```

```
\columnratio{.64}
\RequirePackage{framed}
\setlength \FrameSep {\z@}
\setlength \parindent {\z@}
```

Config the appearance of appendix in toc.

```
\renewcommand \@pnumwidth {2.5em}
\renewcommand \appendix {
  \cleardoublepage
  \setcounter{chapter}{0}
  \setcounter{section}{0}
  \gdef\@chapapp{\appendixname}
  \gdef\thechapter{\@Alph\c@chapter}
  \addtocontents {toc} { \protect \makeatletter }
  \addtocontents {toc} {
    \token_to_str:N \renewcommand* \token_to_str:N \l@section
      { \token_to_str:N \@dottedtocline {1} {1.5em} {5em} }
  }
  \addtocontents {toc} { \protect \makeatother }
  \renewcommand*\familydefault {\sfdefault}
  \pagenumbering{roman}
}
```

End the input stream of `whats-layout-module.code.tex`.

```
\file_input_stop:
```

End the optionlist `class` for l3docstrip.

```
⟨/layout⟩
```

A.4 The `whats-theorem-module.code.tex` file

Start the optionlist `theorem` for l3docstrip.

```
⟨*theorem⟩
```

Declare the module file.

```
\__whats_provide_module:n { theorem }
```

Load the keytheorems package, and predefine the following environments.

```
\RequirePackage{keytheorems}
\newcounter {week}
\newcommand \newweek { \refstepcounter{week} \clearpage }
\newkeytheorem { theorem    } [ parent    = section, style = definition ]
\newkeytheorem { example    } [ parent    = section, style = definition ]
\newkeytheorem { definition } [ parent    = section, style = definition ]
\newkeytheorem { problem    } [ parent    = week,    style = definition ]
\newkeytheorem { solution   } [ numbered  = false,   style = definition ]
\newkeytheorem { remark     } [ numbered  = false,   style = remark     ]
```

```
\file_input_stop:
```

```
⟨/theorem⟩
```

A.5 The `whats-cover-module.code.tex` file

Start the optionlist `cover` for l3docstrip.

```
⟨*cover⟩
```

Declare the module file.

```
\__whats_provide_module:n { cover }
```

Load the tikz package for graphing the (The `expl3` syntax needs to be temporarily turned off).

```
\RequirePackage{tikz, zref-totpages}
\ExplSyntaxOff
\usetikzlibrary{fadings, patterns, arrows.meta}
\tikzset{ > = Stealth, every picture/.append style = {
          line join = round, line cap = round, thick } }
\ExplSyntaxOn
```

`\coverset` Macro for cover configuration by setting keys.

```
\DeclareDocumentCommand \coverset { m } { \keys_set:nn { __whats / coverset } {#1} }
```

Comma list for storing the left logos' path.

`\l__whats_cover_llogos_clist`

```
\clist_new:N \l__whats_cover_llogos_clist
```

Token lists for storing elements on the cover page.

```
\keys_define:nn { __whats / coverset }
  {
    title         .tl_gset:N    = \@title,
    author        .tl_gset:N    = \@author,
    afill         .tl_gset:N    = \@afill,
    date          .tl_gset:N    = \@date,
      date        .initial:n    = \today,
    extinfo       .tl_set:N     = \l__whats_cover_extinfo_tl,
    head          .tl_set:N     = \l__whats_cover_head_tl,
    clogo         .tl_set:N     = \l__whats_cover_clogo_tl,
    llogos        .clist_set:N  = \l__whats_cover_llogos_clist,
  }
```

`\@title` `\@author` `\@afill` `\@date` `\l__whats_cover_extinfo_tl` `\l__whats_cover_head_tl` `\l__whats_cover_clogo_tl`

(*End of definition for* `\coverset` *and others. These functions are documented on page* 1.)

`\sidehead` `\sidefoot` Macros for configuring the header and footer on the sidebar.

```
\DeclareRobustCommand * \sidehead [1] { \tl_gset:Nn \@sidehead {#1} }
\DeclareRobustCommand * \sidefoot [1] { \tl_gset:Nn \@sidefoot {#1} }
```

`\@sidehead` `\@sidefoot` Set the header and footer of the sidebar ⟨*title name*⟩, ⟨*page number*⟩ / ⟨*total page*⟩ by default.

```
\sidehead{\@title}
\sidefoot{\thepage\,/\,\ztotpages}
```

(*End of definition for* `\sidehead` *and others. These functions are documented on page* 2.)

Declare a hook for sidebar.

```
\hook_gset_rule:nnnn { shipout / background } { shipout__whats }
                     { before } { pgfrcs }
```

\maketitle Redefine the \maketitle macro for graphing the cover and load the hook.

```
\DeclareDocumentCommand \maketitle { O { gray } }
  {
    \__whats_@maketitle:n {#1}
    \hook_gput_code:nnn { shipout / background } { shipout__whats }
      { \if@mainmatter \__whats_sidebar_hook:n {#1} \fi }
  }
```

(*End of definition for* \maketitle. *This function is documented on page* 1.)

__whats_@maketitle:n The auxiliary command for \maketitle.

```
\cs_new_protected_nopar:Npn \__whats_@maketitle:n #1
  {
    \pagecolor{#1!10}
    \let \cleardoublepage \clearpage
    \begin{titlepage}
      \begin{tikzpicture} [ remember ~ picture, overlay ]
        \tl_if_empty:NF \l__whats_cover_head_tl
          {
            \node [ yshift = .325\paperheight, opacity = .5,
                    rotate ~ around = { pi \c_colon_str
                    ([yshift = .325\paperheight]current ~ page.center)} ]
              at (current ~ page.center)
              { \includegraphics [width = \paperwidth]
                { \l__whats_cover_head_tl } };
          }
        \clist_if_empty:NF \l__whats_cover_llogos_clist
          {
            \int_step_inline:nn { \clist_count:N \l__whats_cover_llogos_clist }
              {
                \node
                  [ opacity = .25, yshift = {
                    -.6\paperheight / 2 /
                      \clist_count:N \l__whats_cover_llogos_clist +
                    ##1 * .6\paperheight /
                      \clist_count:N \l__whats_cover_llogos_clist }
                  ] at ([xshift = .1\paperwidth]current ~ page.south ~ west)
                  {
                    \includegraphics
                      [
                        height  = .6\paperheight/6,
                        width   = .16\paperwidth, keepaspectratio
                      ] { \clist_item:Nn \l__whats_cover_llogos_clist { ##1 } }
                  };
              }
          }
        \fill [ white, opacity = .5 ] (current ~ page.south ~ east) rectangle +
          (-.8\paperwidth, .6\paperheight);
        \node [ gray, above ~ left, inner ~ sep = 1em ] at
          (current ~ page.south ~ east) { \l__whats_cover_extinfo_tl };
        \fill [ white, opacity = .5 ] (current ~ page.south ~ east) rectangle +
          (-.7\paperwidth, .7\paperheight);
        \node
          [ shift = {(-.35*\paperwidth,.65*\paperheight)},
```

```
        darkgray, font = \Huge \bfseries \sffamily
      ] at (current ~ page.south ~ east) { \@title };
    \node
      [ shift = {(-.75*\paperwidth,.3*\paperheight)},
      rotate = 90, gray, opacity = .6, font = \huge \sffamily
      ] at (current ~ page.south ~ east) { \@author\ | \ \@afill };
    \tl_if_empty:NF { \l__whats_cover_clogo_tl }
      {
        \node (cover ~ logo)
          [ shift = {(.65*\paperwidth,.3*\paperheight)}]
          at (current ~ page.south ~ west)
          {
            \includegraphics[ height = .4\paperheight ]
              { \l__whats_cover_clogo_tl }
          };
      }
  \end{tikzpicture}
  \clearpage \thispagestyle{empty}
  \centering
  \vspace* \fill \sffamily
    \Large This ~ page ~ was ~ intentionally ~ left ~ blank.
  \vspace* \fill
\end{titlepage}
\pagecolor{Beige!30}
}
```

(*End of definition for* `\__whats_@maketitle:n`.)

`\__whats_sidebar_hook:n` Hook for graphing the sidebar: different on odd and even pages.

```
\cs_new_protected:Npn \__whats_sidebar_hook:n #1
  {
    \int_if_odd:nTF { \arabic { page } }
      {
        \tikz [remember ~ picture, overlay] {
          \fill [ #1, opacity = .125 ]
                (current ~ page.south ~ west) rectangle
                ([xshift = +.5in]current ~ page.north ~ west);
          \node [ opacity = .25, inner ~ xsep = .25in,
                  above, inner ~ ysep = {1in - \footskip + 1ex},
                  font = \sffamily \large,
                ] at ([xshift = +.25in]current ~ page.south ~ west)
                { \rotatebox {-90} {\@sidehead} };
          \node [ opacity = .25, inner ~ xsep = .25in,
                  above, inner ~ ysep = \headsep, yshift = -1in,
                  font = \sffamily \large,
                ] at ([xshift = +.25in]current ~ page.north ~ west)
                { \rotatebox {-90} {\@sidefoot} };
        }
      }
      {
        \tikz [remember ~ picture, overlay] {
          \fill [ #1, opacity = .125 ]
                (current ~ page.south ~ east) rectangle
                ([xshift = -.5in]current ~ page.north ~ east);
```

```
            \node [ opacity = .25, inner ~ xsep = .25in,
                    above, inner ~ ysep = {1in - \footskip + 1ex},
                    font = \sffamily \large,
                  ] at ([xshift = -.25in]current ~ page.south ~ east)
                  { \rotatebox {+90} {\@sidehead} };
            \node [ opacity = .25, inner ~ xsep = .25in,
                    above, inner ~ ysep = \headsep, yshift = -1in,
                    font = \sffamily \large,
                  ] at ([xshift = -.25in]current ~ page.north ~ east)
                  { \rotatebox {+90} {\@sidefoot} };
            }
        }
    }
```

(*End of definition for* `\__whats_sidebar_hook:n`.)

End the input stream of `whats-cover-module.code.tex`.

```
\file_input_stop:
```

End the optionlist `cover` for l3docstrip.

⟨/cover⟩

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